Bearing Component Generator (Version: 2025.2 (Build 292293000, 293))

5/28/2025

Project Info (iProperties)

Guide

Type of Strength Calculation - Check Calculation

Loads

Bearing radial load	F_r	519 N
Bearing axial load	F_a	0 N
Speed	n	1496 rpm
Required static safety factor	s_0	2.0 ul

Bearing

Designation		BS 290 (SKF 6310)
Bearing inside diameter	d	50.000 mm
Bearing outside diameter	D	110.000 mm
Bearing width	B	27.000 mm
Nominal contact angle of the bearing	α	5 deg
Basic dynamic load rating	C	65000 N
Basic static load rating	C_0	38000 N
Dynamic radial load Factor	X	1.00 ul / 0.56 ul
Dynamic axial load Factor	Y	0.00 ul / 0.00 ul
Limit value of F_a/F_r	e	0.40 ul
Static radial load Factor	X_0	0.60 ul
Static axial load Factor	Y_0	0.60 ul
Limiting speed lubrication grease	n_{Lim1}	6300 rpm
Limiting speed lubrication oil	n_{Lim2}	7500 rpm

Bearing Life Calculation

Calculation Method		ANSI/AFBMA 9-1990 (ISO 281-1990)
Required rating life	L_{req}	10000 hr
Required reliability	R_{req}	90 ul
Life adjustment factor for special bearing properties	a_2	1.00 ul
Life adjustment factor for operating conditions	a_3	1.00 ul
Working temperature	T	100 c
Factor of Additional Forces	f_d	1.00 ul

Lubrication

Friction factor	μ	0.0015 ul
Lubrication		Grease

Results

Basic rating life	L_{10}	21856499 hr
Adjusted rating life	L_{na}	21856499 hr
Calculated static safety factor	s_{0c}	73.18952 ul
Power lost by friction	P_z	3.05069 W
Necessary minimum load	F_{min}	380 N
Static equivalent load	P_0	519 N
Dynamic equivalent load	P	519 N
Over-revolving factor	k_n	4.211 ul
Life adjustment factor for reliability	a_1	1.00 ul
Temperature factor	f_t	1.00 ul
Equivalent speed	n_e	1496 rpm
Minimum speed	n_{min}	1496 rpm
Maximum speed	n_{max}	1496 rpm
Strength Check	**Positive**	

Summary of Messages

7:37:49 AM Calculation: Calculation indicates design compliance!